Electron Spin Resonance and Nuclear Magnetic Resonance

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Abstract

The values reported here are representative data, generated to demonstrate the analysis. They are not a record of measurements taken by the author. See the note immediately below.

Electron spin resonance (ESR) and nuclear magnetic resonance (NMR) were used to measure the magnetic moments of an unpaired electron and of two spin one-half nuclei. A Helmholtz pair was calibrated against a Hall magnetometer and gave a coil constant of $(4.2073 \pm 0.0084)\,\mathrm{mT\,A^{-1}}$ (statistical; $\pm 0.043\,\mathrm{mT\,A^{-1}}$ once the 1 % field standard is included), which is 0.57 % below the value of $4.2314\,\mathrm{mT\,A^{-1}}$ predicted by the Helmholtz formula for 320 turns per coil. Resonance in solid DPPH was located at 24 frequencies between 15 MHz and 130 MHz. The resonance frequency was linear in field with a gradient of frequency against field of $(27.963 \pm 0.071)\,\mathrm{MHz\,mT^{-1}}$ (statistical, with the coil-constant calibration included), giving an electron spin g-factor of $g_s = 1.998 \pm 0.005\,(\mathrm{stat}) \pm 0.020\,(\mathrm{syst}) = 1.998 \pm 0.021$. This is 0.22 % below the free-electron value $g_e = 2.0023$ and 0.29 % below the accepted solid-DPPH value 2.0036 ± 0.0002, an agreement at the level of 0.22 and 0.28 standard deviations respectively. The absorption line was $(0.238 \pm 0.018)\,\mathrm{mT}$ wide at half maximum, implying a transverse relaxation time of $(47.8 \pm 3.6)\,\mathrm{ns}$, against 22 to 44 ns implied by the published peak-to-peak widths. An iron-cored electromagnet was calibrated over 3.0 to 4.0 A; its gradient fell by 20.6 % across that range and the rising and falling branches differed by 5.2 mT at 3.00 A. Proton resonance in glycerine gave a gradient of frequency against field of $(42.48 \pm 0.38)\,\mathrm{MHz\,T^{-1}}$ and $g(^1\mathrm{H}) = 5.57 \pm 0.08$, against a literature value of 5.5857 (0.17 standard deviations). Fluorine resonance in PTFE gave $(39.56 \pm 0.69)\,\mathrm{MHz\,T^{-1}}$ and $g(^{19}\mathrm{F}) = 5.19 \pm 0.10$, against 5.2577 (0.66 standard deviations). Polythene was found to resonate at $(17.80 \pm 0.05)\,\mathrm{MHz}$ at 0.4194 T, which is about 0.3 standard deviations from the proton prediction and about 5 standard deviations from the fluorine prediction once the 1 % uncertainty on the calibrated field is carried, identifying it as a hydrocarbon containing no fluorine.

Note on the data in this report.
The original laboratory notebook for this experiment is no longer available. Every measured value in this report, and every table and figure derived from measurements, is therefore **representative**: they were generated from accepted physical constants plus realistic instrument scatter, in order to demonstrate the analysis, the uncertainty treatment and the physics of the experiment. **None of it is a record of measurements taken by the author.** The apparatus, the procedure, the theory and the literature comparisons are genuine and follow the departmental experiment information script. The measurements themselves are not. Nothing in this document should be read, cited or reused as experimental evidence.

1 Introduction

Magnetic resonance is the absorption of electromagnetic radiation by a magnetic moment placed in a static magnetic field. The moment has a set of discrete energy levels whose spacing is proportional to the field, and when an applied photon matches that spacing, transitions are driven and energy is absorbed. The absorption is sharp in frequency, so the field at which it occurs can be located precisely, which is what makes the effect useful as a measurement.

Two versions of the same physics are studied here. In electron spin resonance (ESR), also called electron paramagnetic resonance, the moment belongs to an unpaired electron. In nuclear magnetic resonance (NMR) it belongs to an atomic nucleus. The two differ only in the size of the moment. The nuclear magneton is smaller than the Bohr magneton by the ratio of the proton mass to the electron mass, about 1836, but the proton g-factor is 5.586 against 2.002 for the electron, so the resonance field needed at a given frequency is larger by $g_s\mu_B/(g_p\mu_n) = 658$, not by 1836 [2]. That single number explains almost every difference between the two halves of the apparatus: ESR here uses air-cored Helmholtz coils producing a few millitesla, while NMR needs an iron-cored electromagnet producing about 0.42 T.

Both techniques matter practically. ESR is limited to species with a permanent electronic magnetic moment, so it is used on free radicals, transition-metal ions, colour centres and conduction electrons, and the measured g-factor reports directly on the orbital environment of the unpaired electron [3]. NMR determines molecular structure and bonding across chemistry and materials science, and is the physical basis of magnetic resonance imaging [4, 5]. The organic free radical 2,2-diphenyl-1-picrylhydrazyl, referred to throughout by its script name diphenyl-picryl-hydrazyl (DPPH), is used for the ESR half: it is stable indefinitely as a solid at room temperature, carries one unpaired electron per molecule and gives a single narrow line, which is why it is the standard ESR reference material [6]. The NMR half uses glycerine and polystyrene as sources of protons, PTFE as a source of fluorine, and polythene as an unknown to be identified from its resonance frequency.

The experiment has six objectives, taken from the departmental script [1]. For ESR: graph the resonance frequency as a function of magnetic field; determine the g-factor of DPPH; and determine the line width of the resonance signal. For NMR: investigate the resonance of protons in glycerine and polystyrene; investigate the resonance of fluorine in PTFE; and determine the g-factors of the proton and of fluorine. The script also asks for the composition of polythene to be deduced from its resonance frequency, and for the transients known as wiggles to be observed and explained.

2 Theory

2.1 Magnetic moments and the two magnetons

An electron of charge $-e$ and mass m_e moving in a circular orbit of radius ρ at speed v is a current loop of current $i = -ev/2\pi\rho$, so its magnetic moment is

$$\mu_L = iA = -\frac{ev}{2\pi\rho}\,\pi\rho^2 = -\frac{e}{2m_e}\,(m_e v\rho) = -\frac{e}{2m_e}L = -\frac{e\hbar}{2m_e}\,m_\ell \equiv -\mu_B m_\ell, \qquad \mu_B = \frac{e\hbar}{2m_e}, \quad (1)$$

where $L = m_\ell\hbar$ is the orbital angular momentum and $\mu_B = 9.274\,010\,065\,7 \times 10^{-24}\,\mathrm{J\,T^{-1}}$ is the Bohr magneton, the script's equation (2). Orbital motion therefore has a g-factor of exactly 1. Spin does not: the Dirac equation gives $g_s = 2$ for a point spin one-half particle, and quantum electrodynamic corrections raise this slightly. The general relation covering both contributions is the script's equation (1),

$$\boldsymbol{\mu}_J = -g_J\,\mu_B\,\frac{\mathbf{J}}{\hbar}, \quad (2)$$

where $\mathbf{J}$ is the total electronic angular momentum and g_J is the Lande splitting factor, given for a state of well-defined L, S and J by $g_J = 1 + [J(J+1) + S(S+1) - L(L+1)]/[2J(J+1)]$. For a pure spin state, $L = 0$ and $J = S = 1/2$, so $g_J = 2 = g_s$. This is the case identified in the script: for DPPH the orbital contribution is quenched and only the spin need be considered.

For a nucleus the same construction applies with the proton mass in place of the electron mass, giving the nuclear magneton $\mu_n = e\hbar/2m_p = 5.050\,783\,739\,3 \times 10^{-27}\,\mathrm{J\,T^{-1}}$, so that $\mu_B/\mu_n = m_p/m_e = 1836.2$. The nuclear moment is written, following the script's equation (8), as

$$\boldsymbol{\mu} = g\,\mu_n\,\frac{\mathbf{S}}{\hbar}, \quad (3)$$

where $\mathbf{S}$ is the nuclear spin and g the nuclear g-factor.[1] Unlike the electronic case, g for a composite nucleus cannot be calculated from first principles. The proton has $g = 5.586$ and the neutron

[1] The script uses S for the nuclear spin quantum number and k for its projection, where most of the literature uses

$g = -3.826$, and those anomalous values were among the first clear indications that neither particle is elementary.

2.2 Zeeman splitting and the resonance condition

The potential energy of a magnetic moment in a field $\mathbf{B} = B_0\hat{z}$ is the script's equation (3), $E = -\boldsymbol{\mu}\cdot\mathbf{B}$. Substituting equation (2) gives $E = g_J\mu_B m_J B_0$, and for the free electron $m_J \to m_s = \pm 1/2$, which is the script's equation (4),

$$E_\pm = \pm\tfrac{1}{2}\, g_s\, \mu_B\, B_0, \qquad m_s = \pm\tfrac{1}{2}, \tag{4}$$

corresponding to the electron magnetic moment being antiparallel and parallel to the field respectively, that is to the spin being parallel and antiparallel to it. Because the moment carries a minus sign relative to the spin, the lower state is the one in which the moment is parallel to $\mathbf{B}$. The splitting is $\Delta E = g_s\mu_B B_0$.

An oscillating field B_1 applied perpendicular to B_0 drives magnetic dipole transitions with the selection rule $\Delta m_s = \pm 1$. Absorption occurs when the photon energy matches the splitting, which is the script's equation (5):

$$h\nu = g_s\, \mu_B\, B_0. \tag{5}$$

Rearranged for the quantity the experiment extracts from a graph,

$$\boxed{g_s = \frac{h}{\mu_B}\left(\frac{\mathrm{d}\nu}{\mathrm{d}B_0}\right)}, \qquad \frac{h}{\mu_B} = 7.144\,774 \times 10^{-11}\,\mathrm{T\,s}. \tag{6}$$

The whole of ESR objectives (i) and (ii) reduces to measuring that gradient. Note that the estimator is the gradient and not a single pair of readings. A constant offset in the field, for example from the Earth's field, shifts the intercept and leaves the gradient untouched, so taking the gradient makes the result immune to it. This point is used in Sections 4.2 and 5.

The nuclear case is algebraically identical. With $2S + 1$ equally spaced levels

$$E_k = -g\,\mu_n B_0\, k, \qquad k = -S,\, -S+1,\, \ldots,\, +S, \tag{7}$$

which is the script's equation (9), the selection rule $\Delta k = \pm 1$ gives the script's equation (11),

$$h\nu = |E_{k+1} - E_k| = g\,\mu_n B_0, \qquad \boxed{g = \frac{h}{\mu_n}\left(\frac{\mathrm{d}\nu}{\mathrm{d}B_0}\right)}, \qquad \frac{h}{\mu_n} = 1.311\,889 \times 10^{-7}\,\mathrm{T\,s}. \tag{8}$$

Both ^{1}H and ^{19}F have $S = 1/2$, so both have exactly two levels and a single resonance line. Equal level spacing is what makes the spectrum a single line rather than a multiplet: every allowed step on the ladder has the same energy.

The accepted gradients follow directly from the constants. For the free electron, $g_e\mu_B/h = 28.0250\,\mathrm{MHz\,mT^{-1}}$; for DPPH at $g = 2.0036$, $28.0429\,\mathrm{MHz\,mT^{-1}}$; for the proton, $g_p\mu_n/h = 42.577\,478\,\mathrm{MHz\,T^{-1}}$; and for ^{19}F, $g_F\mu_n/h = 40.077\,583\,\mathrm{MHz\,T^{-1}}$ [2, 8, 9].

2.3 Why DPPH, and why the g-factor sits close to two

Equation (5) describes something observable only if the sample has a non-zero electronic magnetic moment in its ground state. In a closed-shell molecule every orbital is doubly occupied with paired spins, the total electronic angular momentum vanishes, and there is no splitting to detect. A paramagnetic species is one with a permanent non-zero total electronic angular momentum, which in practice means an unpaired electron. DPPH is a stable free radical whose unpaired electron sits on the hydrazyl bridge and is delocalised over the two hydrazyl nitrogen atoms and out into the surrounding rings.

Two effects fix its g-factor. First, orbital quenching. DPPH is a low-symmetry organic molecule with an orbitally non-degenerate ground state whose wavefunction can be taken real, and for a

I and m_I. The script's symbols are kept throughout this report. Both $\mathbf{S}$ and $\mathbf{J}$ carry units of angular momentum, so both moment definitions divide by $\hbar$. The symbol k_B always means the Boltzmann constant.

real wavefunction $\langle\psi|\hat{L}_i|\psi\rangle = 0$, because $\hat{L}_i$ is purely imaginary in the position representation while the expectation value of a Hermitian operator must be real. Physically, the electron occupies a standing-wave molecular orbital held in place by the bonding framework rather than a circulating one, so the orbital contribution is quenched to first order and $g_J \to g_s \approx 2$.

Second, a small residual shift, because quenching is not complete. Spin-orbit coupling mixes a little excited-state orbital character into the ground state, and second-order perturbation theory gives

$$\Delta g_{ii} = g_{ii} - g_e = -2\lambda\,\Lambda_{ii}, \qquad \Lambda_{ii} = \sum_{n\neq 0} \frac{\left|\langle 0|\hat{L}_i|n\rangle\right|^2}{E_n - E_0}, \tag{9}$$

which is a tensor; a single principal component is quoted here. Every term of the sum is positive, so the sign of Δg_{ii} is the opposite of the sign of the effective spin-orbit constant. The sign is positive for DPPH because the spin density sits on the hydrazyl nitrogens, where the relevant excitations are from filled orbitals, so the effective spin-orbit constant is negative and Δg comes out positive. For a light-atom organic radical the spin-orbit constant λ is small in magnitude and the relevant excitation energies are of order several electronvolts, so Δg is only about $+1.3 \times 10^{-3}$, giving $g(\mathrm{DPPH}) = 2.0036 \pm 0.0002$, about 0.065 % above g_e [6]. That shift is a chemical fingerprint, and reading it is the whole basis of ESR as a structural technique. Section 6 returns to whether the present apparatus can resolve it.

2.4 Population differences and signal strength

The absorption signal exists only because the lower level is slightly more populated than the upper one. The populations follow the script's equation (10), the Boltzmann distribution, and for a two-level system the fractional excess in the lower level is

$$P = \frac{N_+ - N_-}{N_+ + N_-} = \tanh\!\left(\frac{\Delta E}{2k_B T}\right) \simeq \frac{\Delta E}{2k_B T}, \tag{10}$$

the approximation being excellent because $\Delta E/k_B T \sim 10^{-6}$ here. At the laboratory temperature of 295 K recorded in Section 3, DPPH at $B_0 = 2.0$ mT gives $P = 4.6$ ppm, about five spins in a million, and protons at 0.43 T give $P = 1.49$ ppm. Two comparisons follow and are used in the discussion. At equal field the electron splitting exceeds the proton splitting by $g_s\mu_B/g_p\mu_n = 658.2$. For induction-detected magnetic resonance the signal scales as $\gamma^3 B_0^2$, because the induced electromotive force carries an extra factor of the precession frequency, so at equal field the electron advantage is $658.2^3 = 2.85 \times 10^8$. That $\gamma^3 B_0^2$ scaling is the standard figure of merit for induction detection and is quoted here only for orientation. The present rig detects the loss in a marginal oscillator rather than an induced electromotive force, as Section 3.2 describes, so its signal is proportional to $\eta\chi'' Q$ and carries an additional dependence on T_2; the exact power of γ therefore differs, though the order of magnitude of the advantage does not. A commercial X-band spectrometer at 0.34 T and the same temperature achieves $P = 776$ ppm, a factor of 170 better than the present rig.

2.5 The Helmholtz field

A single circular loop of radius r carrying current I produces an on-axis field $B(z) = \mu_0 I r^2/[2(r^2 + z^2)^{3/2}]$ at axial distance z. Two identical coaxial coils of n turns each, separated by $d = r$, give at the midpoint

$$B_0 = 2 \times \frac{\mu_0 n I r^2}{2\,(r^2 + (r/2)^2)^{3/2}} = \left(\frac{4}{5}\right)^{3/2} \frac{\mu_0\, n\, I}{r}, \tag{11}$$

which is the script's equation (6). The first derivative vanishes at the midpoint by symmetry for any separation. The separation $d = r$ is chosen because it also makes the second derivative vanish, and the third vanishes by symmetry, so the leading non-uniformity is fourth order, leaving $B(z) = B_0[1 - (144/125)(z/r)^4 + \ldots]$. Over a 6 mm sample at $r = 68$ mm the on-axis fractional variation is only 4.36×10^{-6}, which is negligible.

Evaluating with $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{V\,s\,A^{-1}\,m^{-1}}$, $n = 320$ and $r = 6.8\,\mathrm{cm}$ gives $(4/5)^{3/2} = 0.71554$, $\mu_0 n/r = 5.913\,586 \times 10^{-3}\,\mathrm{T\,A^{-1}}$ and therefore $B_0 = 4.2314\,\mathrm{mT\,A^{-1}} \times I$. The script states this simplifies to $B_0 = 4.23I$ in millitesla, which is confirmed to three significant figures. It nevertheless instructs that the measured calibration be used in preference, since the coils may not have exactly the stated number of turns. That instruction is followed here and equation (11) is used only as a comparison.

2.6 Line width and relaxation

Following the script, ΔB_0 denotes the *full width at half maximum* (FWHM) of the absorption line. The script calls this the half-width, but the procedure it prescribes measures the width at half the full signal height, which is the FWHM. The distinction matters by a factor of two in the relaxation time and by $\sqrt{3}$ in the comparison with published widths, so it is fixed here and used consistently. For a homogeneously broadened line the absorption is Lorentzian, $\chi''(\omega) \propto T_2/[1 + (\omega - \omega_0)^2 T_2^2]$, whose half-width at half maximum in angular frequency is $1/T_2$. Converting to field units through $\omega = \gamma B$ with $\gamma = g\mu_B/\hbar$,

$$\Delta B_0^{\mathrm{FWHM}} = \frac{2}{\gamma T_2} = \frac{2\hbar}{g\mu_B T_2}, \qquad \Longrightarrow \qquad T_2 = \frac{2\hbar}{g\,\mu_B\,\Delta B_0}. \tag{12}$$

Commercial spectrometers use lock-in detection and record the first derivative of the absorption, so the literature quotes the peak-to-peak derivative width ΔB_{pp}. For a Lorentzian the derivative extrema sit at $\pm\Gamma/\sqrt{3}$, where Γ is the half-width at half maximum, giving

$$\Delta B_{pp} = \frac{\Delta B_0^{\mathrm{FWHM}}}{\sqrt{3}}. \tag{13}$$

This conversion is applied before any comparison with published values is made.

Several mechanisms broaden the DPPH line. A rigid lattice of dipoles at the spacing found in solid DPPH would give a width of order 4.3 mT, and the unresolved hyperfine structure from the two hydrazyl nitrogens would add of order 1 mT. The line observed in the solid is far narrower than either, because Heisenberg exchange between neighbouring radicals motionally averages the local dipolar field. The Anderson-Weiss result for exchange narrowing is $\Delta B_{\mathrm{obs}} \sim (\Delta B_{\mathrm{dip}})^2/\Delta B_{\mathrm{ex}}$, and it also predicts that the surviving line shape is Lorentzian with extended wings rather than Gaussian [7]. The same mechanism collapses the nitrogen hyperfine quintet seen in dilute solution into the single line seen in the solid.

For the NMR samples the controlling parameters are the relaxation times: T_1 governs the recovery of the longitudinal magnetisation and therefore how easily the transition saturates, T_2 governs the loss of transverse phase coherence, and the observed decay time follows $1/T_2^{*} = 1/T_2 + \gamma\,\Delta B_0^{\mathrm{sample}}$, which adds dephasing from inhomogeneity of B_0 across the sample. In a liquid such as glycerine the molecules reorient far faster than the dipolar dephasing time, so the proton-proton coupling averages towards zero and the line is narrow. In a glass such as polystyrene, well below its glass transition, the backbone does not move on the resonance timescale, the full rigid-lattice coupling survives, and the line is broad.

2.7 Transient oscillations after passage

The script asks for the transients known as wiggles to be observed and explained. Work in a frame rotating at the fixed radio frequency and let $t = 0$ be the instant the swept field passes exactly through resonance. Transverse magnetisation created at that instant precesses freely while the field continues to move, so the offset grows linearly, $\Delta\omega(t) = \gamma(\mathrm{d}B/\mathrm{d}t)\,t$, the accumulated phase is quadratic, and the detected in-phase component behaves as

$$\phi(t) = \tfrac{1}{2}\gamma\left(\frac{\mathrm{d}B}{\mathrm{d}t}\right)t^2, \qquad S(t) \propto \exp\left(-t/T_2^{*}\right)\cos\!\left[\tfrac{1}{2}\gamma\,(\mathrm{d}B/\mathrm{d}t)\,t^2\right], \tag{14}$$

a chirped, decaying oscillation whose successive maxima fall at $t_n = \sqrt{4\pi n/\gamma|\mathrm{d}B/\mathrm{d}t|}$. The spacing therefore shrinks as $\sqrt{n+1} - \sqrt{n}$: the wiggles crowd closer together as they die away, and they only

ever trail the resonance, never precede it, because they are the free decay of magnetisation created at the moment of passage. The number of maxima that appear before the envelope has decayed is

$$N = \frac{\gamma \, (\mathrm{d}B/\mathrm{d}t) \, (T_2^*)^2}{4\pi}, \tag{15}$$

so *one* countable wiggle requires $\gamma(\mathrm{d}B/\mathrm{d}t)(T_2^*)^2 > 4\pi$ and n of them require n times that. For a Lorentzian line the decay time is $T_2^* = 1/\pi\,\Delta\nu$, with $\Delta\nu$ the line width in frequency units. The modulation used here was about 2.4 mT peak at 28 Hz, so $\mathrm{d}B/\mathrm{d}t = 0.42\,\mathrm{T\,s^{-1}}$ at the zero crossing, and four to six wiggles then need T_2^* of order a millisecond, hence a line no more than a few hundred hertz wide. Their presence is thus a direct qualitative proof that the magnet is homogeneous and the sample properly centred, and their number is a rough measure of T_2^*. The counts observed are compared with equation (15) in Section 4.6 and Figure 8. The effect was seen in the first nuclear induction experiments [10] and analysed shortly afterwards [11].

3 Apparatus and method

Both halves of the experiment were run on the open bench at the ambient laboratory temperature of 295 K, about 22 °C. Nothing in either procedure is temperature controlled and no sample was heated or cooled, so the temperature enters the calculations in only one place: it sets the Boltzmann population difference that makes the absorption detectable at all. Every population calculation in this report, for the electron and for the proton alike, uses this value. It matters qualitatively in one further place, since it fixes where each polymer sample sits relative to its own glass transition, which is what Section 4.6 uses to explain the line widths.

3.1 ESR apparatus

The ESR assembly comprised an ESR basic unit carrying the radio-frequency oscillator, an ESR control unit, a pair of Helmholtz coils and a two-channel oscilloscope. The whole assembly was clamped as shown in Appendix 3 of the script, because mechanical instability spoils the sharpness of the resonance by modulating the position of the sample in both the static field and the radio-frequency coil.

The Helmholtz coils were positioned mechanically parallel at a separation of 6.8 cm, equal to the coil radius, since the separation fixes the field configuration. They were connected electrically in series, the E and E sockets joined and the A sockets taken to the current supply of the ESR control unit. The basic unit was connected to the control unit through the six-pole cable at the socket labelled PROBE, the ESR SIGNAL output taken to input CH II of the oscilloscope and the B SIGNAL output to input CH I, with both channels displayed and time-synchronised. Three interchangeable plug-in coils cover the frequency range: the largest 15 to 30 MHz, the medium coil 30 to 75 MHz and the smallest 75 to 130 MHz. The DPPH sample, held in a small glass tube, was inserted into the centre of whichever coil was in use, and that coil placed at the centre of the Helmholtz pair.

3.2 The marginal oscillator detection scheme

The plug-in coil containing the sample is the inductor of the tank circuit of an oscillator biased to run only just above the threshold for sustained oscillation. This is what *marginal* means: the loop gain barely exceeds unity, so the oscillation amplitude is extremely sensitive to any additional loss in the tank. The radio-frequency current in the coil produces a linearly polarised field B_1 along the coil axis, perpendicular to the Helmholtz axis, as the transition requires.

At resonance the sample presents a complex magnetic susceptibility $\chi = \chi' - i\chi''$. If the empty-coil inductance is L_0 and the sample occupies a fraction η of the coil volume, the loaded inductance is $L = L_0(1 + \eta\chi' - i\eta\chi'')$ and the coil impedance becomes

$$Z = i\omega L = i\omega L_0(1 + \eta\chi') + \underbrace{\omega L_0 \, \eta \, \chi''}_{\Delta R}. \tag{16}$$

Two things happen at once. The dispersive part χ' shifts the effective inductance and pulls the tank frequency slightly. The absorptive part χ'' adds a real dissipative resistance ΔR in series with the coil, reducing the quality factor by $\Delta Q/Q = -\eta\chi''Q$. Because the oscillator sits at the margin, that small extra loss produces a disproportionately large drop in the radio-frequency voltage across the coil. That drop is the signal: it is envelope-detected and amplified inside the basic unit and appears at the ESR SIGNAL output. The displayed quantity is therefore proportional to $\chi''(B)$, the absorption line shape itself and not its derivative, which matters for the width measurement of Section 3.6. The same detection principle, due originally to Pound and Knight [12], is used in the NMR half. Radio-frequency amplitude must not be pushed too hard, because a large B_1 equalises the level populations and thereby flattens and broadens the line; the script's instructions to keep the signal symmetric and the modulation small are instructions to stay out of saturation and out of over-modulation respectively.

3.3 Field modulation and the two oscilloscope displays

A 50 Hz modulation current $I_{\sim}$ was superimposed on the steady Helmholtz current $I_{=}$, so that the field at the sample was

$$B(t) = B_0 + B_{\mathrm{mod}} \sin\left(2\pi f_{\mathrm{mod}} t\right), \qquad f_{\mathrm{mod}} = 50\,\mathrm{Hz}. \tag{17}$$

The modulation is a slow sweep, not a small dither. Its purpose is to carry the field back and forth through the resonance condition one hundred times per second so that the absorption line appears as a stationary repetitive trace on an ordinary oscilloscope.

Because the radio frequency is fixed, resonance occurs whenever $B(t) = B_{\mathrm{res}} = h\nu/(g_s\mu_B)$. A sine passes any value in its range exactly twice per period, once rising and once falling, so two absorption pulses appear per cycle. Writing $\epsilon = (B_{\mathrm{res}} - B_0)/B_{\mathrm{mod}}$, the two crossings sit at phases $\arcsin\epsilon$ and $\pi - \arcsin\epsilon$, separated by $\Delta\phi = \pi - 2\arcsin\epsilon$. When $B_0 = B_{\mathrm{res}}$ the pulses are separated by exactly half a modulation period and the train on the screen is evenly spaced; otherwise they bunch to one side. Both cases are drawn in Figure 1(a). Adjusting $I_{=}$ until the signals are equally spaced is therefore a null method for setting B_0 to the resonance field, and it is far more sensitive than judging when a single peak is centred. This was the criterion used at every frequency.

In XY mode the oscilloscope plots the ESR signal on Y against the B SIGNAL on X, so the horizontal axis becomes a field axis and the trace is the absorption line shape against field. The two traversals per cycle should retrace the same curve but do not, because the actual field at the sample lags the reference: the coils are inductive and the detector chain adds its own delay. A phase error φ separates the two traces horizontally by an amount proportional to $2B_{\mathrm{mod}}\sin\varphi$, opening the trace into a loop. The PHASE potentiometer applies a compensating shift so that the two traces coincide, as in Figure 1(b). Coincidence is the condition under which the X axis is a faithful linear field axis, so it was set before any width was read. Equal spacing in dual mode and coincidence in XY mode are two views of the same correct tuning, and the oscilloscope was toggled between them to confirm both.

3.4 Calibration of the Helmholtz coils

Following Appendix 4 of the script, the magnetometer was switched to HIGH FIELD and zeroed with the PROBE ZERO control. The Hall probe was inserted into the standard magnet, stated as 112 mT with a tolerance of 1 %, and the display set to that value with PROBE CAL. The probe was then moved to the centre of the Helmholtz pair, the instrument switched to LOW FIELD, and the display zeroed again with the current off, which subtracts whatever ambient field lies along the probe axis at that point, including the Earth's field. The Helmholtz current was then set from 0.1 A to 1.2 A in 0.1 A steps and the field read at each.

3.5 ESR measurement procedure

With the largest plug-in coil fitted and the DPPH sample in place, the frequency was set to 15 MHz, the modulation current $I_{\sim}$ set to mid-position and the PHASE potentiometer turned fully to the

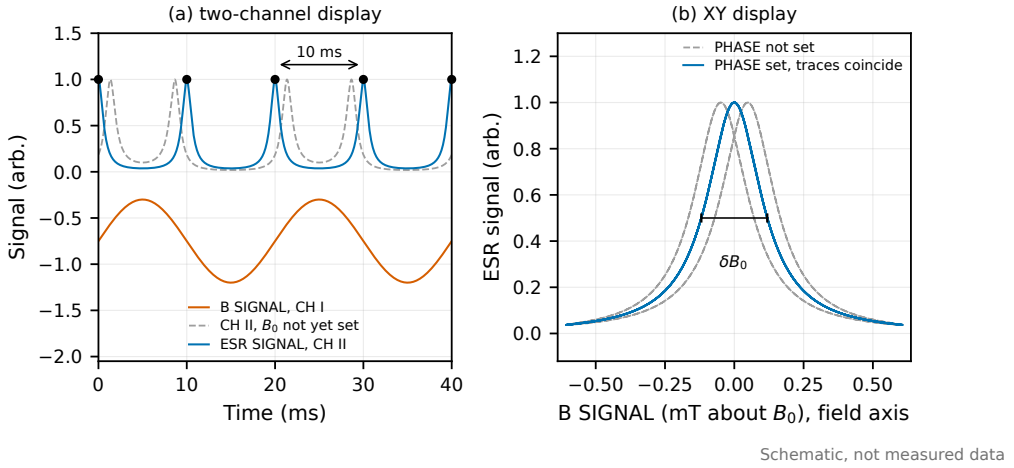


Figure 1: Schematic of the two ESR oscilloscope displays. Both panels are drawn from the representative parameters of the 50 MHz line-width measurement: $B_0 = 1.740\,\mathrm{mT}$, a modulation swing of 1.214 mT peak to peak and a line 0.238 mT wide at half height. (a) The two-channel display, with the B SIGNAL on CH I and the ESR SIGNAL on CH II. On resonance the two absorptions per modulation cycle are 10 ms apart, half a modulation period, and that even spacing is the null condition used to set $I_{=}$; the grey trace is the same display with B_0 set 0.25 mT low, where the signals bunch into pairs. (b) The XY display, absorption against the field axis, with the PHASE control set so that the two traversals coincide and the half-height width δB_0 can be read; the grey loop is the same signal with an uncompensated lag of 0.08 rad. This is a schematic of the instrument display, not a captured trace and not measured data.

right. The control unit display was toggled to A= and the steady current raised slowly until the resonance signals were equally spaced. The oscilloscope was then switched to XY mode, the phase adjusted until the two signals coincided, and the display returned to dual mode. The steady current was trimmed for a symmetrical signal and the modulation current reduced as far as the signal would allow.

The steady current $I_{=}$ and the frequency ν were recorded, the frequency increased in 5 MHz steps and the current readjusted at each. The measurement was repeated five times at every frequency so that the scatter could be assessed, as the script requires. The procedure was then repeated with the medium coil over 35 to 75 MHz and the smallest coil over 80 to 130 MHz, giving 24 frequencies. The script warns that sharp resonance cannot be obtained at every frequency and that some waveforms appear distorted. In this data set a symmetric signal was obtainable at every 5 MHz step, so no frequency had to be discarded and all 24 points are retained. Each frequency was taken on one plug-in coil only, so although the three bands overlap, no frequency was measured twice on two different coils.

3.6 Line-width procedure

The resonance condition was established at 50 MHz on the medium coil and the oscilloscope switched to XY mode with both channels at $0.5\,\mathrm{V\,cm^{-1}}$. The modulation current was increased until the trace spanned exactly the 10.0 cm width of the screen, and that root-mean-square current recorded as I_{mod}, the corresponding full modulation voltage being $U_{\mathrm{mod}} = 0.5\,\mathrm{V\,cm^{-1}} \times 10.0\,\mathrm{cm} = 5.00\,\mathrm{V}$. The X deflection was then expanded to $0.2\,\mathrm{V\,cm^{-1}}$ and the width δU of the signal at half its full height read in volts. Note that δU is a voltage and is therefore independent of the volts-per-centimetre setting used to read it; expanding the trace magnifies it by a factor of 2.5 for easier reading but does not change δU.

Since the full 10 cm of X deflection corresponds to the peak-to-peak modulation current, and the root-mean-square value must be multiplied by $2\sqrt{2}$ to obtain that, the proportion carries straight across and gives the script's equation (7),

$$\frac{\delta I}{2\sqrt{2}\,I_{\mathrm{mod}}} = \frac{\delta U}{U_{\mathrm{mod}}} \qquad \Longrightarrow \qquad \delta I = 2\sqrt{2}\;I_{\mathrm{mod}}\,\frac{\delta U}{U_{\mathrm{mod}}}, \tag{18}$$

after which the width in field units follows from the measured coil constant s as $\Delta B_0 = s\,\delta I$.

3.7 NMR apparatus and procedure

The NMR assembly comprised an NMR supply unit containing a marginal oscillator working between about 15 MHz and 20 MHz, an NMR probe, a pair of electromagnet coils with an iron yoke, separate modulation coils, a DC power supply and the oscilloscope. The electromagnet coils were connected in series to the DC supply and the modulation coils in series with the MODULATION COILS output. The probe was connected to the PROBE input, the DB SIGNAL output taken to CH I of the oscilloscope and the NMR SIGNAL output to CH II, both AC coupled, with the oscilloscope placed several inches from the magnet to avoid distortion of the display.

The HF AMPLITUDE control was set to minimum and FREQUENCY to maximum, FAST SWEEP was selected and the modulation amplitude set to maximum. The glycerine sample tube was inserted into the measuring chamber and its O-ring adjusted so that the sample sat at the chamber centre. The unit was switched on and the HF amplitude raised slowly until a frequency of about 19 MHz was displayed, then reduced to about 18.5 MHz. With the DC supply controls at minimum, the coil current was brought to about 3.2 A and the frequency decreased slowly until resonance signals appeared. The signal was optimised by varying the HF amplitude, then centred by adjusting the coil current or the frequency while reducing the modulation amplitude. The oscilloscope was switched to XY mode and the PHASE of the modulation voltage adjusted so that the signals from both half-waves coincided, then returned to dual mode. The modulation frequency is 28 Hz, not a harmonic or sub-harmonic of the 50 Hz mains supply, so that, whatever the reason for the choice, mains pickup drifts across the trace instead of locking to it and imitating signal. Both displays are drawn in Figure 8.

The procedure was repeated for polystyrene, for polythene and for PTFE, and the transients described in Section 2.7 recorded for each sample. Polythene and polystyrene were then examined at a single fixed current only, because their lines were too broad and weak to locate reliably at every set point; the full frequency-against-current run of Section 3.8 was taken for glycerine and for PTFE.

3.8 Electromagnet calibration and the one-directional sweep

For the quantitative part, the apparatus was first tuned on glycerine, the sample removed and the Hall probe of the teslameter inserted carefully into the measuring chamber. The field was measured against the series coil current for an increasing sweep from 3 A to 4 A and then for a decreasing sweep back to 3 A, in 0.1 A steps.

The two branches do not coincide, because the yoke is ferromagnetic. Magnetisation proceeds by the motion of domain walls, which are pinned by grain boundaries, dislocations and inclusions. Unpinning requires a finite driving field and the energy released when a wall breaks free is dissipated, so the process is irreversible and B lags behind H [13]. Sweeping the current from 3 A to 4 A and back traces a minor loop rather than the full major loop, because the current never reverses and never returns to zero, so the two branches must converge at the upper turning point, and the gap is largest near the lower one.

The consequence is procedural and is stated by the script: the current must be moved in one direction only. The calibration is single-valued on one branch only, and the subsequent frequency run converts each current to a field using it. Backing the current off and re-approaching lands the magnet on a different minor loop, and that error is systematic rather than random and one-signed, so it does not average out with repetition.

The glycerine sample was returned to the chamber and the resonance frequency measured against current from 3.00 A to 4.00 A in 0.10 A steps, the current raised monotonically throughout. The measurement was repeated for PTFE, and finally all four samples were compared at the single fixed current of 3.50 A.

4 Results and analysis

All fits in this section are least squares. The Helmholtz and electromagnet calibrations carry uncertainty on the ordinate only, so the standard closed-form weighted expressions of Section 5.2 apply, with weights $w_i = 1/\sigma_i^2$. The ESR resonance fit and both NMR frequency fits carry uncertainty

on both axes, and the larger part of it sits on the abscissa, so both use the effective-variance method of Section 5.2, iterated to convergence, with the field uncertainty projected onto the frequency axis through the gradient. Reduced chi-squared is quoted for every fit.

4.1 Calibration of the Helmholtz coils

Twelve field readings were taken from 0.1 A to 1.2 A, covering 0.42 to 5.03 mT. The full data are in Table A1. A straight line $B = sI + B_{\mathrm{off}}$ was fitted with $\sigma_B = 0.010\,\mathrm{mT}$ on every point. The coil current is treated as exact in this fit. The same ammeter reading defines the current in the calibration and in the resonance run, so a scale or offset error in it is common to both and cancels in the ratio that gives the g-factor; only its random component matters, and that is already captured in the repeat scatter of the resonance currents. The fit is shown in Figure 2 and the results in Table 1.

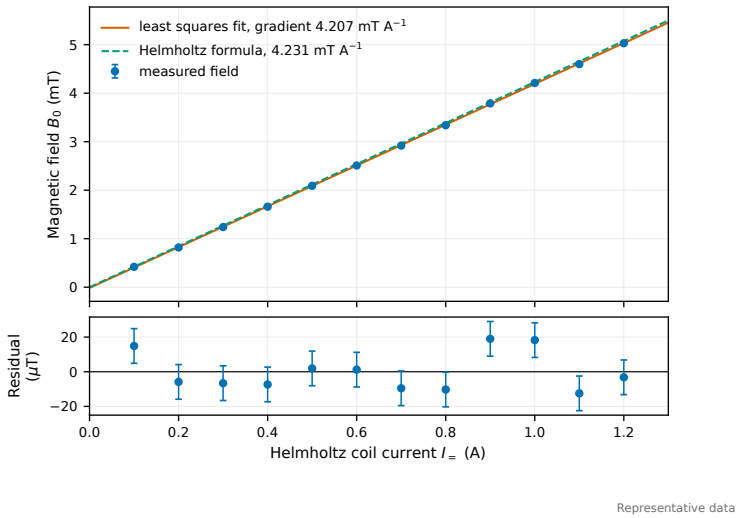


Figure 2: Helmholtz coil calibration from representative data. Upper panel: measured field at the coil midpoint against steady coil current, with the least squares fit and, for comparison, the theoretical line of equation (11). Lower panel: fit residuals in microtesla. The largest residual is 19.0 µT, about twice the assigned reading uncertainty.

Table 1: Helmholtz calibration fit, from representative data.

Quantity	Value	Note
Coil constant s, statistical only	$(4.2073 \pm 0.0084)\,\mathrm{mT\,A^{-1}}$	from the fit
Coil constant s, statistical and systematic	$(4.207 \pm 0.043)\,\mathrm{mT\,A^{-1}}$	includes 1 % reference magnet
Intercept B_{off}	$(-0.0156 \pm 0.0062)\,\mathrm{mT}$	residual zeroing offset
Chi-squared	14.15	10 degrees of freedom
Reduced chi-squared	1.41	
Largest residual	19.0 µT	
Theoretical value, equation (11)	$4.2314\,\mathrm{mT\,A^{-1}}$	$n = 320$, $r = 6.8\,\mathrm{cm}$
Difference from theory	$-0.0241\,\mathrm{mT\,A^{-1}}$, $-0.57\,\%$	
Effective turns per coil implied	318.2 ± 3.2	stated value 320, consistent

The fitted coil constant is 0.57 % below the value predicted by the Helmholtz formula for 320 turns. Inverting equation (11) for the turn count gives an effective 318.2 ± 3.2 turns per coil, where the uncertainty is the 1.02 % on the systematic-inclusive coil constant. The gap from 320 is therefore only 0.6 standard deviations and the data cannot distinguish 318 from 320. That is a plausible difference for a hand-wound coil, and it is exactly why the script instructs that the calibration rather than the formula be used. The measured value is used for every field in the ESR analysis. The small negative intercept is consistent with a residual offset on the LOW FIELD zero; it shifts the field axis but not its scale.

4.2 Resonance frequency against magnetic field, and the g-factor of DPPH

Resonance was located at 24 frequencies from 15 MHz to 130 MHz across the three plug-in coils, with five repeat settings of the current at each. The currents ran from 0.117 A to 1.099 A, comfortably

inside the range over which the coils were calibrated, and the corresponding fields from 0.477 mT to 4.607 mT. The mean standard error on a current was 0.0024 A. The full data are in Table A2. The fit is shown in Figure 3 and the results in Table 2.

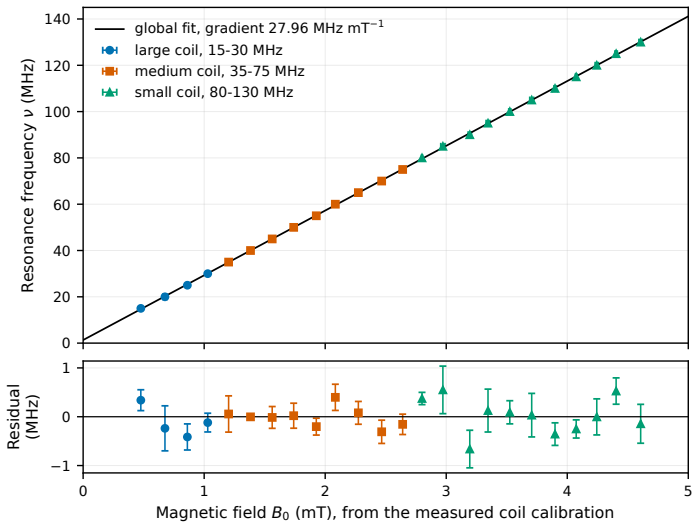


Figure 3: ESR resonance frequency against magnetic field for DPPH, from representative data. Currents have been converted to field with the measured coil constant of Table 1. The three plug-in coils are distinguished by symbol. The line is a single global weighted fit to all 24 points. The lower panel shows the residuals, which scatter about zero and show no offset between the three coil bands beyond what the point scatter allows, although no frequency was measured on two coils, so a common offset smaller than about 0.1 MHz would not have been detected.

Table 2: ESR resonance fit and the extracted g-factor of DPPH, from representative data.

Quantity	Value	Note
Gradient $\mathrm{d}\nu/\mathrm{d}B_0$	(27.963 ± 0.045) MHz mT^{-1}	fit only, calibration uncertainty not included
Gradient, quoted result	(27.963 ± 0.071) MHz mT^{-1}	statistical, calibration included
Intercept ν_0	(1.333 ± 0.114) MHz	
Chi-squared, reduced chi-squared	26.18, 1.19	22 degrees of freedom
Largest residual	0.662 MHz	
Gradient $\mathrm{d}\nu/\mathrm{d}I_=$	(117.65 ± 0.19) MHz A^{-1}	consistency check on the error propagation
k/s from the two gradients	(27.96 ± 0.07) MHz mT^{-1}	agrees identically, as it must
g_s(DPPH), central value	1.9979	equation (6)
Statistical uncertainty	0.0051	fit and calibration combined
Systematic uncertainty	0.0200	1 % reference magnet
Quoted result	$\mathbf{1.998 \pm 0.005}$ (stat) $\pm$ **0.020** (syst) $= \mathbf{1.998 \pm 0.021}$	1.03 %
Literature g(DPPH), and difference	2.0036 ± 0.0002; -0.29 %, 0.28σ	[6]
Free electron g_e, and difference	2.002319304; -0.22 %, 0.22σ	[2]

The plot is linear over an eightfold range of frequency and a nearly tenfold range of field. Both the field and the frequency carry uncertainty, the field through the repeat scatter of the resonance current and the frequency through the 0.1 MHz readout resolution and short-term drift of the oscillator, so the effective-variance method is used. The frequency term is not negligible: it is comparable to the projected field error at the low-field end, and omitting it would understate the gradient uncertainty by about a fifth and inflate the reduced chi-squared from 1.19 to 1.50. Applying equation (6) with

$h/\mu_B = 7.144\,774 \times 10^{-11}$ T s gives $g_s = 1.9979$. Combining the statistical uncertainty of 0.0051 with the 1 % systematic from the reference magnet gives a total of 0.0206, so the result is quoted as

$$g_s(\mathrm{DPPH}) = 1.998 \pm 0.005\,(\mathrm{stat}) \pm 0.020\,(\mathrm{syst}) = 1.998 \pm 0.021. \tag{19}$$

A useful cross-check is to fit ν against current directly and divide by the coil constant. That gives (117.65 ± 0.19) MHz A^{-1} and hence the same 27.963 MHz mT^{-1}, as it must algebraically, but with a larger uncertainty of 0.071 MHz mT^{-1} because the calibration error is then carried explicitly, whereas the direct fit of ν against B_0 uses field uncertainties built only from the current scatter and so omits it entirely. The two routes are not independent determinations; the second is the one that carries the full statistical error and it is the one propagated into g. The direct fit supplies the central value, and the two-gradient route supplies the uncertainty, so the 0.0051 statistical uncertainty on g_s comes from the latter.

The fitted intercept deserves attention because the physics requires it to be zero: an $S = 1/2$ system has no zero-field splitting, since the term responsible vanishes identically for a doublet. The measured intercept of (1.333 ± 0.114) MHz is 12 standard deviations from zero, so it is not consistent with zero. Dividing by the gradient converts it into a field. Relative to the calibrated axis the excess at the DPPH sample is (0.048 ± 0.004) mT; once the calibration's own zero offset of -0.016 mT is removed, the implied ambient field along the coil axis is 0.032 mT. That is the right size for the residual ambient field left after zeroing the magnetometer at the coil centre. Only the axial component matters, and in Glasgow the Earth's field of about 48 µT is steeply inclined, so a horizontal coil axis sees roughly 17 µT of it; the balance is stray field from nearby equipment. An offset of this kind shifts the intercept and leaves the gradient untouched, which is precisely why the g-factor is taken from the gradient. The intercept is therefore reported as a diagnostic rather than forced to zero.

4.3 Line width of the DPPH resonance

The width was measured at 50 MHz on the medium plug-in coil, where the resonance current was 0.417 A and the field (1.7397 ± 0.0092) mT. The measured quantities and the values derived from equation (18) are in Table 3, and the line shape with the half-height construction drawn on it is in Figure 4.

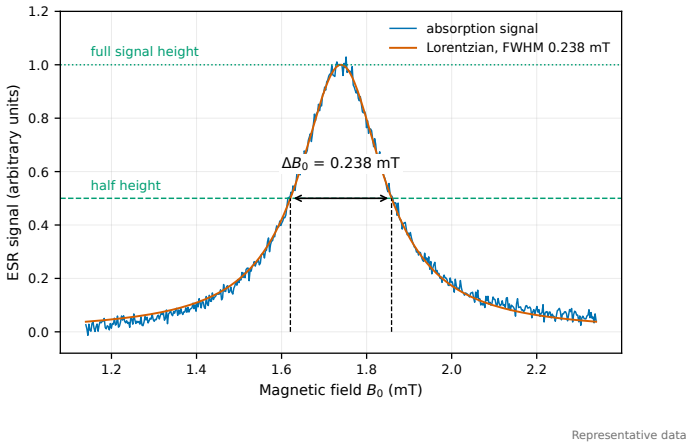


Figure 4: DPPH absorption line at 50 MHz with the half-height width construction, from representative data. The horizontal axis is the field axis provided by the oscilloscope XY display once the PHASE control has been set for coincidence of the two half-wave traces. The smooth curve is a Lorentzian of full width 0.238 mT at half maximum.

Three internal consistency checks were made on this measurement, all of which the data pass. The peak-to-peak modulation field, $s \times 2\sqrt{2} \times I_{\mathrm{mod}} = 1.214$ mT, stays below the steady field of 1.7397 mT, so the field never reverses and the two-crossings-per-cycle picture holds. The line occupies 20 % of the 10 cm trace, that is 2.0 cm at 0.5 V cm^{-1}, expanding to 4.9 cm at 0.2 V cm^{-1}. Both are comfortably readable on a real screen.

Table 3: Line-width measurement at 50 MHz, from representative data.

Quantity	Value	Uncertainty	Note
RF frequency	50.0 MHz	0.1 MHz	set value, medium coil
Resonance current $I_{=}$	0.417 A	0.0022 A	mean of five repeats
Resonance field B_0	1.7397 mT	0.0092 mT	from the calibration
Modulation current I_{mod} (RMS)	0.102 A	0.005 A	trace spans 10.0 cm
Full modulation voltage U_{mod}	5.00 V	0.10 V	0.5 V cm^{-1} over 10.0 cm
Half-height width δU	0.98 V	0.05 V	read at 0.2 V cm^{-1}
Half-width current δI	0.0565 A	0.0042 A	equation (18)
Line width ΔB_0 (FWHM)	0.238 mT	0.018 mT	7.4 %
Same, in gauss	2.38 G	0.18 G	
Peak-to-peak derivative width	0.137 mT	0.010 mT	equation (13)
Same, in gauss	1.37 G	0.10 G	for literature comparison
Relaxation time T_2	47.8 ns	3.6 ns	equation (12)

Because $\Delta B_0 = s\,\delta I$, the 1 % field-standard systematic scales the line width as well as the g-factors, and it is already carried inside the 7.4 % quoted above. It contributes only 1.0 % of that total, against 5.1 % from δU and 4.9 % from I_{mod}, so the width is limited by reading the trace rather than by the field standard.

Applying equation (12) with the fitted g gives $T_2 = (47.8 \pm 3.6)$ ns, or a relaxation rate of $1/T_2 = 2.090 \times 10^{7}\ \mathrm{s}^{-1}$. This conversion assumes the line is homogeneously broadened and Lorentzian. Any residual inhomogeneous broadening means the measured width gives T_2^{*} rather than T_2, so the quoted figure is properly a lower bound on the true transverse relaxation time.

4.4 Calibration of the electromagnet and its hysteresis

Eleven points were taken on each branch from 3.00 A to 4.00 A. The full data are in Table A3. The rising branch was fitted with a quadratic $B(I) = a_0 + a_1 I + a_2 I^2$ by effective-variance weighted least squares, the current uncertainty being projected onto the field axis through $\mathrm{d}B/\mathrm{d}I$. Results are in Table 4 and the fit in Figure 5.

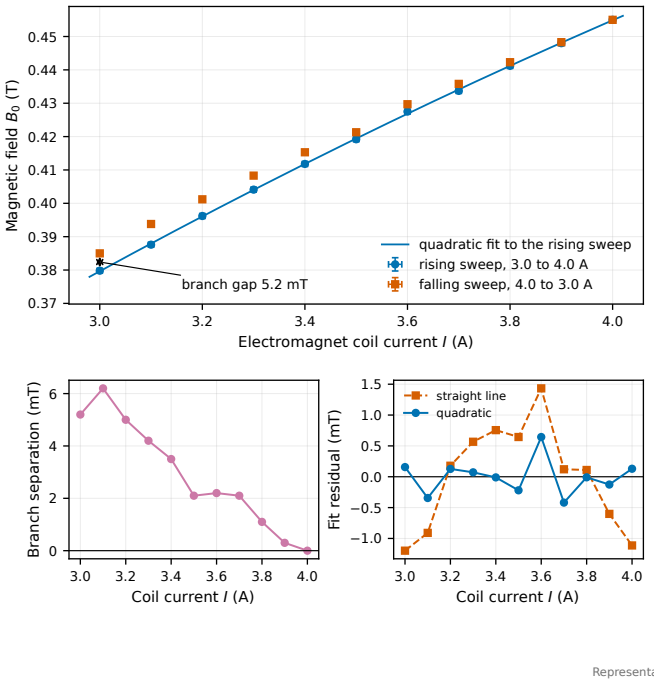


Figure 5: Electromagnet calibration from representative data. Upper panel: both sweep branches with the quadratic fitted to the rising branch, and the branch gap at 3.00 A marked. Lower left: the separation between the two branches, which is largest near the lower end of the sweep and closes at the upper turning point. Lower right: residuals to a straight line and to the quadratic. The straight-line residuals show the systematic curvature that justifies the higher order.

Three points follow. First, the order of the polynomial is justified rather than assumed. The

Table 4: Electromagnet calibration on the rising branch, and the model comparison, from representative data.

Quantity	Value	Note
a_0	(0.050 ± 0.026) T	not a remanent field, see text
a_1	(0.136 ± 0.015) T A^{-1}	
a_2	(-0.0087 ± 0.0021) T A^{-2}	4.0σ from zero
Chi-squared, quadratic	2.15	8 degrees of freedom
Reduced chi-squared, quadratic	0.27	
Largest quadratic residual	0.64 mT	
Reduced chi-squared, straight line	2.06	9 degrees of freedom
Largest straight-line residual	1.43 mT	against 0.5 mT scatter
Improvement in chi-squared on adding I^2	16.4	one extra parameter, $F = 61.0$
Cubic coefficient a_3	(0.0008 ± 0.0080) T A^{-3}	0.10σ, consistent with zero
Improvement in chi-squared on adding I^3	0.01	$F = 0.03$, not justified
$\mathrm{d}B/\mathrm{d}I$ at 3.00, 3.50, 4.00 A	(0.0839 ± 0.0022), (0.0752 ± 0.0006), (0.0666 ± 0.0022) T A^{-1}	falls by 20.6 %, yoke saturation
Field at 3.00, 3.50, 4.00 A	0.3796 T, 0.4194 T, 0.4549 T	
Branch separation at 3.00, 3.50, 4.00 A	5.2 mT, 2.1 mT, 0.0 mT	1.37 % of B at 3.00 A; branches meet at the top
Equivalent proton frequency shift at 3.00 A	0.221 MHz	11 times the setting precision

magnetic circuit argument requires $\mathrm{d}B/\mathrm{d}I$ to fall with current, so any adequate model must have negative curvature, and a quadratic is the lowest order that can provide it. A straight line leaves residuals with obvious systematic curvature, largest at 1.43 mT against a point scatter of 0.5 mT, and a reduced chi-squared of 2.06. Adding the quadratic term improves chi-squared by 16.4 for one extra parameter, an F statistic of 61.0, while adding a cubic improves it by 0.01, an F statistic of 0.03, with the cubic coefficient itself 0.10σ from zero. The quadratic is the right model and the cubic is not justified.

Second, the reduced chi-squared of 0.27 means the assigned uncertainties are slightly conservative rather than that the fit is suspect: the teslameter scatter of 0.5 mT and the current reading uncertainty of 0.005 A were both assessed before the run rather than inferred from the residuals.

Third, the fitted intercept $a_0 = 0.0501$ T is *not* a remanent field but an extrapolation from 3 A back to 0 A, far outside the range fitted. Real remanence for a magnet of this kind would be 10 to 30 mT, several times smaller. The quadratic is a local empirical interpolation, in effect a Taylor expansion of the true $B(I)$ truncated at second order, and must not be used outside 3.0 to 4.0 A.

The hysteresis is large enough to matter. The branch separation of 5.2 mT at 3.00 A is 1.37 % of the field there and corresponds to a proton resonance shift of 0.221 MHz, eleven times the precision with which the frequency can be set, and it closes to zero at the upper turning point as a minor loop must. The largest separation observed was 6.2 mT at 3.10 A, the 5.2 mT quoted at 3.00 A being one reading of the same quantity, limited by the 0.5 mT reading scatter. This is the quantitative justification for the one-directional sweep rule of Section 3.8.

4.5 The g-factors of the proton and of fluorine

Resonance frequencies were recorded against current for glycerine and for PTFE on the rising branch only, eleven points each. Glycerine gave 16.14 to 19.37 MHz and PTFE 15.28 to 18.31 MHz, both inside the oscillator's stated band. Each current was converted to a field with the quadratic of Section 4.4, carrying the full propagated uncertainty including all three coefficient covariances and the current setting error, which gives a mean field uncertainty of 0.50 mT. The frequency scatter was taken as 0.020 MHz for glycerine and 0.050 MHz for PTFE, set by the display resolution and the short-term drift of the marginal oscillator. The full data are in Table A4. The fits are in Figure 6 and the results in Table 5.

Applying equation (8) gives $g(^1\mathrm{H}) = 5.57 \pm 0.08$ and $g(^{19}\mathrm{F}) = 5.19 \pm 0.10$. Both agree with the

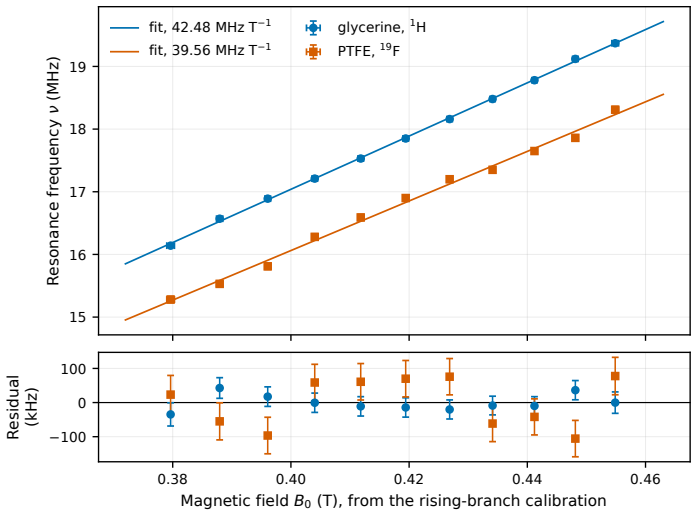


Figure 6: NMR resonance frequency against magnetic field for protons in glycerine and fluorine in PTFE, from representative data. Fields come from the rising-branch quadratic calibration of Figure 5. Lower panel: fit residuals in kilohertz. The PTFE residuals are visibly larger than the glycerine residuals, as the difference in the assumed frequency scatter requires.

Table 5: NMR gradients and g-factors for ^{1}H in glycerine and ^{19}F in PTFE, from representative data.

Quantity	^{1}H in glycerine	^{19}F in PTFE
Frequency range	16.14 to 19.37 MHz	15.28 to 18.31 MHz
Assumed frequency scatter	0.020 MHz	0.050 MHz
Mean propagated field uncertainty	0.50 mT	0.50 mT
Gradient $\mathrm{d}\nu/\mathrm{d}B_0$, route A	(42.48 ± 0.38) MHz T^{-1}	(39.56 ± 0.69) MHz T^{-1}
Relative statistical uncertainty	0.90 %	1.74 %
Intercept c, and its significance	(0.05 ± 0.16) MHz, 0.30σ	(0.24 ± 0.29) MHz, 0.83σ
Implied extra field c/m	(1.1 ± 3.8) mT	(6.1 ± 7.3) mT
Chi-squared, reduced chi-squared (9 dof)	6.22, 0.69	18.62, 2.07
Largest residual	43 kHz	106 kHz
Gradient, route B at 3.50 A	(42.53 ± 0.42) MHz T^{-1}	(39.57 ± 0.71) MHz T^{-1}
$\mathrm{d}\nu/\mathrm{d}I$ at 3.50 A	(3.199 ± 0.019) MHz A^{-1}	(2.976 ± 0.048) MHz A^{-1}
Gradient, Monte Carlo cross-check	(42.51 ± 0.42) MHz T^{-1}	(39.58 ± 0.70) MHz T^{-1}
Literature gradient, and difference	42.5775 MHz T^{-1}, −0.23 %	40.0776 MHz T^{-1}, −1.30 %
g, central value	5.5726	5.1892
Statistical uncertainty	0.0503	0.0904
Systematic uncertainty	0.0557	0.0519
Quoted result	$\mathbf{5.57 \pm 0.08}$	$\mathbf{5.19 \pm 0.10}$
Total relative uncertainty	1.35 %	2.01 %
Literature g	5.585695	5.257736
Difference	−0.23 %, 0.17σ	−1.30 %, 0.66σ

literature values of 5.5857 [2] and 5.2577 [8, 9], at 0.17σ and 0.66σ respectively.

Three ways of propagating the same data and the same calibration were compared, and they agree; they test the error propagation rather than provide independent determinations. The one independent test is the gradient ratio below, in which the field scale cancels. Route A, quoted as the result, fits ν against B_0 over the whole field span using the effective-variance method. Route B, which follows the chain rule the script's wording suggests, fits ν against current directly and divides the two local gradients at 3.50 A. Route B is systematically the more conservative of the two, because it evaluates a local gradient at one current and discards the lever arm of the full range: for the proton it gives 0.99 % against 0.90 %. A Monte Carlo sampling of the calibration covariance over 5000 draws returned (42.51 ± 0.42) MHz T^{-1} for the proton, in agreement with both.

A further test cancels the correlated field scale entirely. The ratio of the two fitted gradients is

0.9312 ± 0.0183, against the literature ratio $40.0776/42.5775 = 0.94129$, a difference of 1.07 % and 0.55σ. Because both gradients were measured against the same calibrated field, the 1 % reference magnet uncertainty divides out of the ratio, so this is a genuinely independent check of the two nuclear moments against each other.

Both intercepts are consistent with zero, at 0.30σ and 0.83σ. Converting them into field offsets gives (1.1 ± 3.8) mT and (6.1 ± 7.3) mT, neither significant. No probe-zero error or remanence offset was therefore detected at the level the data can resolve, which is a few milliteslas.

4.6 Comparison of the four samples and identification of polythene

All four samples were compared at the fixed current of 3.50 A, corresponding to (0.4194 ± 0.0005) T. At that field the two candidate nuclei are predicted to resonate at 17.858 MHz for ^{1}H and 16.809 MHz for ^{19}F, a separation of 1.049 MHz or 5.87 %. Results are in Table 6 and Figure 7.

Table 6: Four-sample comparison at 3.50 A and 0.4194 T, from representative data. The glycerine and PTFE frequencies here are independent settings from those at 3.50 A in Table A4; the differences of 0.02 MHz are one standard deviation of the frequency-setting scatter. The last column gives the overall line width and the number of transients counted; the widths are compared with the transient counts through equation (15) in the text.

Sample	Formula	Nucleus	Observed ν (MHz)	Line width (mT), wiggles
glycerine	$C_3H_5(OH)_3$	^{1}H	17.87 ± 0.02	0.010, 4 to 6 and strong
polystyrene	$(C_8H_8)_n$	^{1}H	17.81 ± 0.12	0.500, none
polythene	$(C_2H_4)_n$	^{1}H (deduced)	17.80 ± 0.05	0.150 composite, 1 to 2 and weak
PTFE	$(C_2F_4)_n$	^{19}F	16.88 ± 0.05	0.150, none
Prediction for ^{1}H at this field			17.858	
Prediction for ^{19}F at this field			16.809	
Separation of the two hypotheses			1.049	5.87 %

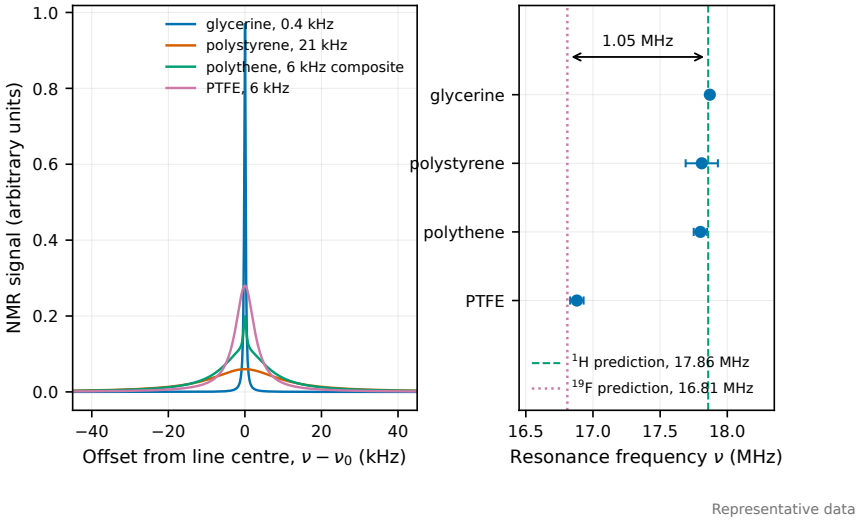


Figure 7: Four-sample comparison at 3.50 A, from representative data. Left: line shapes on a common frequency offset axis, showing the very narrow glycerine line, the broad polystyrene line, and the intermediate PTFE and composite polythene lines. Right: observed resonance frequencies against the two predictions. Glycerine, polystyrene and polythene fall on the proton line and only PTFE falls on the fluorine line.

The polythene line was observed at (17.80 ± 0.05) MHz. It lies 0.058 MHz from the proton prediction and 0.991 MHz from the fluorine prediction. Both predictions are proportional to the calibrated field, which carries the 1 % reference-magnet scale error, that is 0.179 MHz on the proton prediction and 0.168 MHz on the fluorine one, so the honest significances are about 0.3 standard deviations from the proton prediction and about 5 standard deviations from the fluorine one. Both are dominated by that field uncertainty, which shifts the two predictions together. The comparison that removes it is with the observed lines: polythene sits 0.070 MHz from the glycerine line and 0.920 MHz from the PTFE line. Polythene therefore contains hydrogen and contains no fluorine. The handbook formula, $(C_2H_4)_n$ [15], confirms the deduction.

Polystyrene was observed at $(17.81 \pm 0.12)\,\mathrm{MHz}$, which is $0.048\,\mathrm{MHz}$ from the proton prediction, that is 0.4 standard deviations of its own measurement uncertainty. The larger uncertainty reflects the difficulty of centring so broad and weak a line.

Two further observations support it. No second line was found anywhere in the oscillator's range: if polythene contained both nuclei, two resonances separated by more than a megahertz would have appeared. And carbon is silent, because $^{12}\mathrm{C}$ has even proton and neutron numbers and therefore zero spin. The only other candidate, $^{13}\mathrm{C}$, is spin one-half but is present at about 1 % abundance and resonates far below the oscillator's band at this field, so it cannot be confused with anything.

Polythene and PTFE are exact structural analogues, the same backbone with every hydrogen replaced by fluorine, which is what makes them such a clean pair for this comparison. The line shape gives a secondary clue that is consistent with the identification. The polythene line is clearly broader than glycerine's but noticeably narrower than polystyrene's and has a composite appearance, a narrow component on a broad base. Polythene is semicrystalline with a glass transition far below room temperature, so its amorphous fraction is mobile enough to narrow the dipolar coupling while the crystalline lamellae stay rigid. Polystyrene is a glass well below its transition and is uniformly rigid, which is why it shows only the broad component and no wiggles at all.

The two oscilloscope displays and the transient train are drawn in Figure 8, and the wiggle counts in Table 6 follow equation (15) quantitatively. With $\mathrm{d}B/\mathrm{d}t = 0.42\,\mathrm{T\,s^{-1}}$, glycerine's 0.010 mT line gives $T_2^* = 748\,\mu\mathrm{s}$ and $N = 5.0$, which is the four to six observed. Polythene's transients are controlled not by its overall 0.150 mT width but by the narrow amorphous component of about 0.020 mT, giving $T_2^* = 374\,\mu\mathrm{s}$ and $N = 1.3$, the one or two weak ones observed. PTFE at 0.150 mT gives $N = 0.02$ and polystyrene at 0.500 mT gives $N = 0.002$, so neither can show even one, which is what was seen.

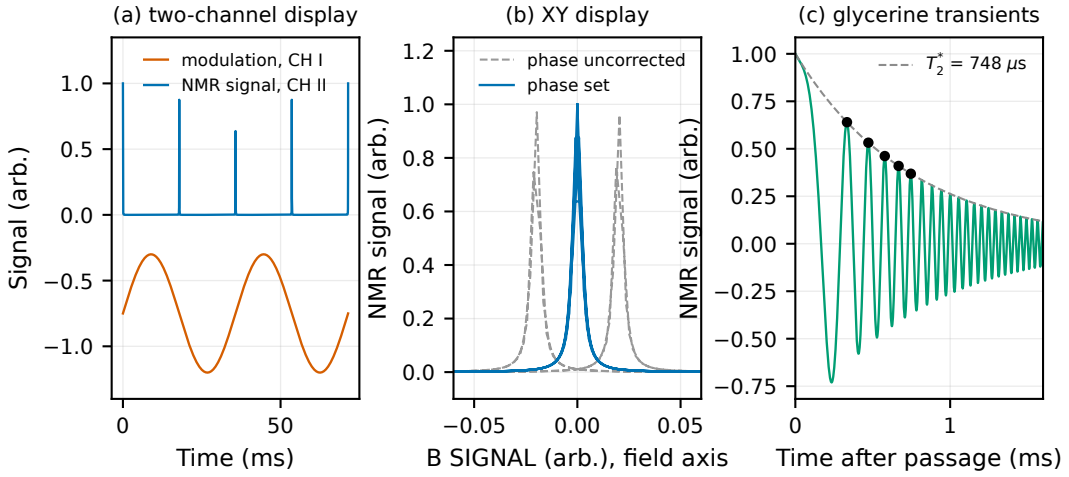


Figure 8: The NMR displays and the transients, from representative data. (a) The two-channel display: the NMR absorption on CH II and the modulation reference on CH I, with two absorptions per modulation cycle, evenly spaced when B_0 is on resonance. (b) The same signal in XY mode with the PHASE control set so that the two half-wave traces coincide, which is the condition under which the X axis is a faithful field axis. (c) The glycerine transient train after passage through resonance, with the successive maxima of $t_n = \sqrt{4\pi n/\gamma|\mathrm{d}B/\mathrm{d}t|}$ marked, showing the predicted crowding as the envelope decays with T_2^*.

5 Uncertainties

5.1 Random uncertainties

Random uncertainties were assessed from the repeat readings where repeats were taken, and from instrument resolution otherwise. Table 7 lists them.

The standard error on the mean resonance current varies from 0.0005 A to 0.0041 A between frequencies, a factor of eight, because the three plug-in coils differ in signal-to-noise and the setting is easier at some frequencies than others. The points therefore do not carry equal information and

Table 7: Random uncertainty contributions, from representative data.

Quantity	Size	Origin
$I_{=}$ at ESR resonance	0.0024 A mean standard error	standard error of five repeats; limited by how finely the eye can judge equal spacing against a line width of 0.238 mT, that is 0.057 A
ESR frequency ν	0.1 MHz	readout resolution and short-term oscillator drift; carried in the fit weights
ESR field in the calibration	0.010 mT	magnetometer display resolution and noise
ESR calibration current $I_{=}$	treated as exact	the same ammeter serves the calibration and the resonance run, so a scale or offset error cancels in k/s; the random part is already in the repeat scatter
δU	0.05 V	reading two edges to about 1 mm on the expanded scale
I_{mod}	0.005 A	control unit display resolution
Full modulation voltage U_{mod}	0.10 V	oscilloscope Y gain accuracy at 0.5 V cm^{-1} over the full 10.0 cm
NMR teslameter reading	0.5 mT	scatter of the reading, display step 0.1 mT
NMR coil current setting	0.005 A	ammeter reading
NMR frequency, glycerine	0.020 MHz	display resolution and short-term drift of the marginal oscillator
NMR frequency, PTFE	0.050 MHz	the same, with the broader line making the null harder to judge

the fit is weighted by $1/\sigma^2$. Had the scatter been constant the weighted fit would have reduced exactly to ordinary least squares.

The frequency uncertainties are set by the display resolution and short-term drift of the oscillators, not by the line widths. The NMR lines are far narrower than the assumed scatters: glycerine's 0.010 mT is only 0.43 kHz and PTFE's 0.150 mT is 6.0 kHz, against assumed scatters of 20 kHz and 50 kHz. A line cannot be the origin of an uncertainty larger than itself.

5.2 Propagation algebra

For a weighted straight-line fit $y = mx + c$ with weights $w_i = 1/\sigma_i^2$, writing $\Delta = (\sum w_i)(\sum w_i x_i^2) - (\sum w_i x_i)^2$, the standard results are

$$m = \frac{(\sum w_i)(\sum w_i x_i y_i) - (\sum w_i x_i)(\sum w_i y_i)}{\Delta}, \qquad \sigma_m^2 = \frac{\sum w_i}{\Delta}, \qquad \sigma_c^2 = \frac{\sum w_i x_i^2}{\Delta}, \tag{20}$$

with $\chi^2 = \sum w_i (y_i - m x_i - c)^2$ and $\chi^2_\nu = \chi^2/(N-2)$ [14].

For the ESR result the measurement passes through two fits in series, exactly as the script warns. With $k = \mathrm{d}\nu/\mathrm{d}I_{=}$ from the resonance fit and s the calibration gradient,

$$\frac{\mathrm{d}\nu}{\mathrm{d}B_0} = \frac{k}{s}, \qquad g_s = \frac{h}{\mu_B}\frac{k}{s}, \qquad \frac{\sigma_g}{g} = \sqrt{\left(\frac{\sigma_k}{k}\right)^2 + \left(\frac{\sigma_s}{s}\right)^2}, \tag{21}$$

the two fits being statistically independent because they come from separate runs. Planck's constant is exact by definition of the SI and the Bohr magneton is known to three parts in 10^{10}, so neither contributes.

For the NMR result the abscissa of the second fit is itself computed from the first, so every point carries a horizontal error and those errors are correlated through the shared fit coefficients. With $\mathbf{J}_i = (1,\, I_i,\, I_i^2)$ and C the covariance matrix of (a_0, a_1, a_2),

$$\sigma_{B_i}^2 = \mathbf{J}_i \, C \, \mathbf{J}_i^{\mathsf{T}} + \left(\left.\frac{\mathrm{d}B}{\mathrm{d}I}\right|_{I_i}\right)^2 \sigma_I^2, \tag{22}$$

where the first term written out is

$$\sigma_{a_0}^2 + I_i^2\sigma_{a_1}^2 + I_i^4\sigma_{a_2}^2 + 2I_i\,\mathrm{cov}(a_0,a_1) + 2I_i^2\,\mathrm{cov}(a_0,a_2) + 2I_i^3\,\mathrm{cov}(a_1,a_2). \tag{23}$$

The cross terms are not optional: polynomial coefficients are strongly anti-correlated and dropping them overestimates σ_B substantially. The second fit then minimises the effective-variance chi-squared,

$$\chi^2 = \sum_i \frac{(\nu_i - mB_i - c)^2}{\sigma_{\nu_i}^2 + m^2 \sigma_{B_i}^2}, \tag{24}$$

in which a horizontal error projects onto the vertical axis through the gradient. Because m appears in the weights the fit is iterated to convergence, which took three passes.

The point-to-point correlation induced by the shared calibration coefficients is neglected in this chi-squared, which sums the per-point variances as if they were independent. Its size is bounded by the mean fractional field uncertainty, 0.12 %, against a statistical gradient uncertainty of 0.90 %. The Monte Carlo cross-check samples the calibration covariance in full and so does carry the correlation; its agreement at $(42.51 \pm 0.42)\,\mathrm{MHz\,T^{-1}}$ is what justifies the approximation.

For the line width all four inputs are independent, so

$$\frac{\sigma_{\Delta B_0}}{\Delta B_0} = \sqrt{\left(\frac{\sigma_{\delta U}}{\delta U}\right)^2 + \left(\frac{\sigma_{I_\mathrm{mod}}}{I_\mathrm{mod}}\right)^2 + \left(\frac{\sigma_{U_\mathrm{mod}}}{U_\mathrm{mod}}\right)^2 + \left(\frac{\sigma_s}{s}\right)^2}, \tag{25}$$

which with the values of Table 3 gives 7.4 %. The line-width measurement is intrinsically far less precise than the g-factor measurement, and the reason is that it depends on reading two edges of a trace by eye rather than on a null condition.

5.3 Systematic uncertainties

The structural point is that almost every systematic here scales the field axis multiplicatively. Since $g \propto 1/s$, such terms feed straight into g as a fractional error and no amount of repetition reduces them. Table 8 lists them with estimated magnitudes and, where the sign is known, its direction.

Only the reference magnet tolerance is carried into the quoted uncertainties. The probe linearity, alignment and positioning terms are bounds rather than measured corrections and are not quantified here; including them at the mid-points of the ranges in Table 8 would raise the total systematic from 1.00 % to about 1.3 %, and the quoted g-factors would become $g_s(\mathrm{DPPH}) = 1.998 \pm 0.027$, $g(^1\mathrm{H}) = 5.57 \pm 0.09$ and $g(^{19}\mathrm{F}) = 5.19 \pm 0.11$. Each agreement then improves in proportion, and it is worth naming the value each one is measured against: DPPH would sit 0.17σ from the free-electron $g_e = 2.0023$ and 0.21σ from the accepted solid-DPPH value 2.0036, the proton 0.15σ from the literature 5.5857, and fluorine 0.61σ from the literature 5.2577. Every uncertainty quoted below and in the abstract is therefore the conservative-in-precision, optimistic-in-accuracy figure that carries the field standard alone.

With that stated, the uncertainty budgets for the three g-factors are as follows. For DPPH, the statistical contribution is 0.26 % and the reference magnet 1.00 %, combining to 1.03 %. For the proton, 0.90 % statistical and 1.00 % systematic combine to 1.35 %. For fluorine, 1.74 % statistical and 1.00 % systematic combine to 2.01 %. In the ESR case the systematic dominates by a factor of four, so taking more repeats would not have improved the result at all. In the fluorine case the statistical term dominates, and there the extra repeats would have helped.

6 Discussion

6.1 The DPPH g-factor and what agreement means

The measured $g_s(\mathrm{DPPH}) = 1.998 \pm 0.021$ agrees with the free-electron value $g_e = 2.0023$ at 0.22σ and with the accepted solid-DPPH value 2.0036 ± 0.0002 at 0.28σ. The central value sits 0.22 % below g_e.

The physically meaningful statement is what this excludes. A purely orbital magnetic moment would have $g = 1$ and a gradient of $\mu_B/h = 13.996\,\mathrm{MHz\,mT^{-1}}$, almost exactly half what was measured, so that hypothesis is excluded by a factor of two, nearly fifty times the total uncertainty. The measurement therefore confirms two things at once: that the orbital angular momentum of the unpaired electron in DPPH is quenched by the molecular bonding framework, and that the

Table 8: Systematic uncertainty budget. Magnitudes marked as estimates are bounds rather than measured corrections. Values quoted for this experiment come from representative data.

Source	Estimated size	Effect
Reference magnet stated as 112 mT with 1 % tolerance	1.0 %	Scales every calibrated field, so scales s, and therefore both g and the line width ΔB_0. The dominant term in every g-factor quoted here. Irreducible without a better field standard.
Hall probe linearity between the calibration point and the working range	0.5 % to 1 %, estimate	The ESR probe is calibrated at 112 mT and used below 5 mT; the NMR probe is used at four times the calibration field. Systematic scaling of s.
Tangential probe misalignment, reading $B\cos\theta$	0.14 % at 3°, 0.38 % at 5°, estimate	Always reads low, never high, so g is biased high. One of the few systematics whose sign is certain.
Probe zeroing offset, and residual ambient field at the ESR sample	up to 0.02 mT; (0.048 ± 0.004) mT measured	Offsets, not scale errors, so they shift the fitted intercept and leave the gradient alone. Neither biases g. The ambient term is about twice the axial component of the Earth's field in Glasgow.
Hysteresis branch error on the electromagnet	5.2 mT at 3.00 A, 1.37 %	The dominant NMR systematic if the sweep rule is broken. One-signed: a field read too high makes g too low. Removed procedurally, not statistically.
Hall probe not at the sample position	0.1 % to 0.5 %, estimate	Probe and sample occupy the chamber at different times; any field gradient then biases B. Acts as a scale error if the geometry is reproducible.
Stated 320 turns per Helmholtz coil	0.57 % discrepancy observed	Bypassed entirely by using the measured calibration. Reported as a property of the coils, not as an error in g.
Coil separation not exactly 6.8 cm	0.88 % per millimetre	Cancels if nothing is moved between calibration and measurement. This is the reason for clamping.
Helmholtz field inhomogeneity over the sample	4.36×10^{-6} fractional	Negligible for g; contributes almost nothing to the line width.
Oscilloscope gain accuracy	already counted	Already carried as the 0.10 V on U_{mod} and the 0.05 V on δU in Table 7. Any gain error common to both partly cancels in the ratio $\delta U/U_{\mathrm{mod}}$, so it is not counted again here.
RF saturation, over-modulation and mechanical instability	broaden the line	Affect ΔB_0 only; the line centre and therefore g are unshifted. Vibration modulates the sample position in field and in the RF coil, which is why the script insists the rig is clamped.
Chemical shielding of the nucleus	0.01 % to 0.06 %, estimate	Biases the NMR g values low. More than twenty times smaller than the experimental uncertainty.

electron's spin g-factor is anomalous, close to two rather than to one. The fitted gyromagnetic ratio is $\gamma = 1.7572 \times 10^{11}\,\mathrm{rad\,s^{-1}\,T^{-1}}$.

What the measurement cannot do is equally worth stating. The DPPH g-shift above the free electron is 0.065 % and the QED anomaly is 0.116 %, while the total uncertainty here is 1.03 %. This experiment therefore cannot resolve g(DPPH) from g_e, and it would be dishonest to claim otherwise. Doing so would need a field standard far better than the 1 % reference magnet, in practice an NMR gaussmeter, which routinely reaches parts per million.

The linearity of the plot is itself a physical result. Straightness over a nearly tenfold range of field, with a reduced chi-squared of 1.19 on 22 degrees of freedom, is consistent with a Zeeman splitting linear in B. The non-zero intercept is attributed to the residual ambient field rather than to a zero-field term, because a zero-field term vanishes identically for a doublet; the data alone cannot separate the two, since a straight-line fit with a free intercept cannot tell a zero-field splitting from a stray field.

6.2 The coil constant and the intercept

The fitted coil constant is 0.57 % below the theoretical $4.2314\,\mathrm{mT\,A^{-1}}$, an effective 318.2 ± 3.2 turns per coil against the stated 320. This tests the turn count, the 6.8 cm separation and the probe calibration together. The difference is not significant against the 1.02 % systematic on the calibration, so it cannot be attributed to any one cause. A turn count below 320, a separation error of about 0.65 mm, since the separation enters at 0.88 % per millimetre, or an error in the probe calibration

would each produce a shift of this size. This is reported as a measured property of the coils rather than as an experimental error, which is what the script intends when it warns that the coils may not have the stated number of turns.

The ESR intercept is genuinely non-zero at 12 standard deviations, corresponding to an extra $(0.048 \pm 0.004)\,\mathrm{mT}$ at the DPPH sample relative to the calibrated axis, or $0.032\,\mathrm{mT}$ of ambient field once the calibration's own $-0.016\,\mathrm{mT}$ zero offset is taken out. The magnetometer was zeroed at the coil centre with the current off, so the calibration has the ambient field subtracted out, but the sample in the actual run still sits in it, and the size is right for the axial component of the Earth's field plus stray field from nearby equipment. The useful point is that the offset shifts the intercept and leaves the gradient untouched, so the g-factor is immune to it. That asymmetry is the strongest single argument for taking the gradient rather than a single pair of readings, and it applies equally to the NMR half, where both intercepts came out consistent with zero.

6.3 Line width, exchange narrowing and relaxation

The measured full width at half maximum is $(0.238 \pm 0.018)\,\mathrm{mT}$, that is $(2.38 \pm 0.18)\,\mathrm{G}$. Comparison with published values requires the conversion of equation (13), because the literature quotes peak-to-peak derivative widths. Converted, the present result corresponds to $(1.37 \pm 0.10)\,\mathrm{G}$ peak-to-peak, against a published range of 1.5 to 3.0 G for polycrystalline DPPH. The measurement sits at or just below the lower edge of that band, 1.2σ from 1.5 G, which is agreement rather than discrepancy. Two reasons for sitting low are worth noting: the oscilloscope reading precision is modest, and published widths for DPPH vary with the sample's preparation and solvent history.

The physics in that number is the comparison with the rigid-lattice estimate of order $4.3\,\mathrm{mT}$, eighteen times wider than what is observed. That factor of eighteen is the signature of exchange narrowing: the radicals in the solid are close enough for the Heisenberg exchange interaction to exceed the dipolar coupling, so each spin samples many local environments in a time short compared with the dipolar dephasing time and the local field averages away [7]. A single width measurement therefore yields a real structural conclusion about the solid.

The transverse relaxation time follows as $(47.8 \pm 3.6)\,\mathrm{ns}$. The published peak-to-peak widths of 1.5 to 3.0 G correspond, through equations (12) and (13), to T_2 between 22 ns and 44 ns, so the present value sits just above that band, consistent with the measured peak-to-peak equivalent of 1.37 G sitting just below it. Any residual inhomogeneous broadening would make the measured width give T_2^{*} rather than T_2, so the quoted number is a lower bound on the true transverse relaxation time.

6.4 The electromagnet, hysteresis and why the sweep rule matters

The contrast between the two halves of the experiment is instructive. The ESR Helmholtz pair is air-cored, so its calibration is a straight line with only a small zero offset of $(-0.0156 \pm 0.0062)\,\mathrm{mT}$, a reduced chi-squared of 1.41 and residuals of a few tens of microtesla. The NMR electromagnet is iron-cored, and its calibration is neither linear nor single-valued.

The nonlinearity is saturation. Applying Ampere's law around the magnetic circuit gives $NI = B l_g/\mu_0 + H_{\mathrm{Fe}}(B)\, l_{\mathrm{Fe}}$. At low fields the iron's permeability is large and the second term is negligible, so B is proportional to I. As the domains align the permeability collapses towards unity, the iron term takes an increasing share of the available ampere-turns, and each extra ampere buys less field. The observed drop of 20.6 % in $\mathrm{d}B/\mathrm{d}I$ across a single ampere is far too large to ignore, and a straight-line calibration would introduce an error of about 0.4 % in B, from a largest straight-line residual of $1.43\,\mathrm{mT}$ on a field of $0.38\,\mathrm{T}$, and therefore the same 0.4 % in g. That is why the script demands a polynomial of order greater than one, and Table 4 confirms that a quadratic is both necessary and sufficient.

The multivaluedness is hysteresis, the dominant potential systematic in the NMR half. The branch separation of $5.2\,\mathrm{mT}$ at $3.00\,\mathrm{A}$ corresponds to a proton frequency shift of $0.221\,\mathrm{MHz}$, eleven times the precision with which the frequency can be set, so it is not a subtle effect. Crucially it is one-signed: overshooting a set point and coming back always leaves the field too high, so ν/B_0 and

therefore g come out too low, and it does not average out with repetition. Both nuclear g-factors did come out slightly low, by 0.23 % and 1.30 %, which is the direction this systematic pushes, although both are well inside their uncertainties and no claim is made that the effect has been detected.

6.5 The nuclear results and the sample comparison

The proton g-factor agrees with the literature at 0.17σ and the fluorine value at 0.66σ. The fluorine measurement is the weaker of the two, with a statistical uncertainty of 1.74 % against 0.90 %, and the reason is visible in the residual panel of Figure 6: the PTFE residuals are two to three times larger. The frequency uncertainties are set by the display resolution and short-term drift of the marginal oscillator, not by the line widths, which are only 0.43 kHz for glycerine and 6.0 kHz for PTFE. What the broader PTFE line does is make the null harder to judge, so the setting scatter is larger there. The ratio test is the cleanest result in the NMR half, because the correlated field-scale uncertainty cancels out of it entirely: the measured 0.9312 ± 0.0183 against the literature 0.94129 is a 0.55σ agreement obtained without needing the reference magnet to be accurate at all, only stable.

The polythene identification is unambiguous by a wide margin. The observed line sits about 0.3 standard deviations from the proton prediction and about 5 from the fluorine prediction once the 1 % field scale is carried on both predictions, and the separation of the two hypotheses is 1.049 MHz, roughly twenty measurement uncertainties and more than a hundred line widths, the widest line in the set, polystyrene's 0.500 mT, corresponding to only 0.021 MHz. The handbook formula $(C_2H_4)_n$ [15] agrees.

The glycerine and polystyrene comparison shows the relaxation physics clearly. Glycerine's observed line is 0.010 mT wide and polystyrene's is 0.500 mT, a factor of fifty. Glycerine is a liquid whose molecules reorient far faster than the dipolar dephasing time, so the proton-proton coupling averages to near zero and the observed width is set by magnet inhomogeneity rather than by relaxation. Polystyrene is a glass well below its glass transition, its backbone frozen on the resonance timescale, so the full rigid-lattice coupling survives. Since peak height goes as integrated intensity divided by width, polystyrene's line is not only broader but very much weaker, which is what makes it hard to find. Its T_1 is also far longer, so at the amplitude that optimises glycerine it is already partly saturated. This is why the script instructs that the HF amplitude be optimised separately for each sample, and it explains the counter-intuitive observation that turning the amplitude up to chase the weak polystyrene signal makes it worse.

The wiggles behave as Section 2.7 predicts, and the counts follow equation (15) quantitatively. Glycerine, with a narrow inhomogeneity-limited line of 0.010 mT, has $T_2^* = 748\,\mu s$ and $N = 5.0$; it showed four to six strong transients. Polystyrene, whose line is fifty times glycerine's, has $T_2^* = 15\,\mu s$ and $N = 0.002$, and showed none. Polythene, whose tabulated width equals PTFE's, showed one or two weak transients while PTFE showed none: polythene's line is a composite, and it is the narrow amorphous component of about 0.020 mT, not the tabulated overall width, that sets T_2^* for the transient, giving $T_2^* = 374\,\mu s$ and $N = 1.3$. PTFE's line is a single homogeneously broadened component with no narrow fraction, so its N is 0.02 and no transient survives. Counting wiggles is therefore a rough but genuine measure of T_2^*, and their presence confirms that the magnet was homogeneous and the sample properly centred.

6.6 Why the apparatus takes the form it does

The Boltzmann numbers explain the design. At 2.0 mT and the laboratory temperature of 295 K the electron population excess is 4.6 ppm, about five spins in a million, and at 0.43 T the proton excess is 1.49 ppm. That is why both halves use a marginal oscillator, whose amplitude is pinned to the point where gain just balances loss and which therefore responds to a tiny extra loss with a large amplitude change; why the ESR rig is clamped; and why the field rather than the frequency is swept, since sweeping a marginal oscillator's frequency would move its working point and swamp the signal.

The electron's advantage over the proton at equal field is 658.2 in splitting. For induction-detected resonance the signal advantage goes as the cube of that, 2.85×10^8, at equal field; the present rig detects the loss in a marginal oscillator rather than an induced electromotive force, so the exact

power differs, but the order of magnitude of the advantage does not. At the fields actually used, 2 mT against 0.42 T, the $\gamma^3 B_0^2$ figure falls to about 6.5×10^3, which is still enough to let ESR run in an air-cored pair while NMR needs iron. Equivalently, the electron polarisation at 2 mT is about three times the proton polarisation at 0.43 T: the field is 215 times smaller but the moment is 658 times larger. Matching the electron polarisation with protons would need 1.3 T, and matching the proton polarisation with electrons needs only 0.65 mT. For context, a commercial X-band spectrometer at 0.34 T achieves 776 ppm, 170 times better than this rig, and adds a resonant cavity on top of that. Low-field ESR is not merely a teaching compromise, though: it is used for in vivo imaging precisely because low-frequency radiation penetrates tissue.

The script warns that sharp resonance cannot be obtained at every frequency and that some waveforms appear distorted. Plausible causes are a plug-in coil operating near the edge of its band, where the oscillator is no longer properly marginal and the tank quality factor is reduced; pickup of mains harmonics in the signal chain; and mechanical resonance of the clamped assembly.

6.7 Improvements

Six concrete improvements follow from the budget in Table 8.

1. **Replace the field standard.** The 1 % tolerance on the reference magnet is the largest single contribution to every g-factor here and is irreducible by repetition. Calibrating against an NMR gaussmeter would reduce it by three or four orders of magnitude and make the statistical term dominant everywhere.
2. **Interleave the NMR runs.** Measure field and frequency at each set point on one continuous upward current sweep, rather than completing the calibration and then the frequency run. This removes almost all of the hysteresis exposure in a single procedural change.
3. **Condition the magnet.** Cycle it two or three times over the full range before calibrating, so the iron settles on a reproducible loop rather than the virgin magnetisation curve, and do not switch it off between the two runs.
4. **Reverse the ESR polarity.** Taking data at both current polarities and averaging cancels the ambient field offset directly rather than merely diagnosing it through the intercept.
5. **Use lock-in detection for the width.** Recording the derivative of the absorption with a small modulation locates the line far more precisely than reading two edges of a trace by eye, and would move the width from 7.4 % towards the g-factor precision.
6. **Fix and measure the geometry.** Mark the Hall probe shaft at the sample O-ring depth and map the field against depth to find the flat region, and measure the Helmholtz separation with callipers. Both convert an estimated systematic into a measured bound.

Two things this apparatus cannot do: it cannot resolve chemical shifts, which needs homogeneity of order one part in 10^9 against the few parts in 10^5 available here, and it cannot measure T_1 or T_2 beyond an order-of-magnitude estimate.

7 Conclusion

The Helmholtz pair was calibrated at $(4.2073 \pm 0.0084)\,\mathrm{mT\,A^{-1}}$ statistical, $\pm 0.043\,\mathrm{mT\,A^{-1}}$ with the 1 % field standard included. That is 0.57 % below the theoretical $4.2314\,\mathrm{mT\,A^{-1}}$, which is 0.6 standard deviations of the systematic-inclusive calibration uncertainty and therefore consistent with the stated 320 turns per coil (318.2 ± 3.2 implied). Resonance in DPPH was linear in field over 15 to 130 MHz with a gradient of $(27.963 \pm 0.071)\,\mathrm{MHz\,mT^{-1}}$ and a reduced chi-squared of 1.19, giving $g_s(\mathrm{DPPH}) = 1.998 \pm 0.021$. That agrees with the free-electron value at 0.22σ and with the accepted DPPH value at 0.28σ, and excludes a purely orbital moment, which would give $g = 1$, by a factor of two. The 1.03 % total uncertainty is more than ten times the 0.065 % DPPH g-shift, so the apparatus cannot resolve that shift. The absorption line was $(0.238 \pm 0.018)\,\mathrm{mT}$ wide at half maximum, eighteen times narrower than the rigid-lattice dipolar estimate, which demonstrates exchange narrowing, and corresponds to $T_2 = (47.8 \pm 3.6)\,\mathrm{ns}$, just above the 22 to 44 ns implied by the published peak-to-peak widths.

The iron-cored electromagnet required a quadratic calibration: its gradient fell by 20.6 % between 3.00 A and 4.00 A, the quadratic coefficient was 4.0σ from zero and a cubic term was not justified at 0.10σ, and the two sweep branches differed by 5.2 mT at 3.00 A. Proton resonance in glycerine gave $(42.48 \pm 0.38)\,\mathrm{MHz\,T^{-1}}$ and $g(^{1}\mathrm{H}) = 5.57 \pm 0.08$, agreeing with 5.5857 at 0.17σ; fluorine in PTFE gave $(39.56 \pm 0.69)\,\mathrm{MHz\,T^{-1}}$ and $g(^{19}\mathrm{F}) = 5.19 \pm 0.10$, agreeing with 5.2577 at 0.66σ. The ratio of the two gradients, 0.9312 ± 0.0183, agrees with the literature ratio at 0.55σ with the correlated field scale cancelled out, and polythene resonated at $(17.80 \pm 0.05)\,\mathrm{MHz}$ at 0.4194 T, about 0.3 standard deviations from the proton prediction and about 5 from the fluorine prediction once the 1 % field scale is carried, identifying it as $(\mathrm{C_2H_4})_n$.

In every case the dominant limitation was the 1 % tolerance on the reference magnet used to calibrate the Hall probe, and the single most valuable improvement would be to replace it with an NMR field standard.

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A Full data tables

All tables in this appendix contain representative data. They are not a record of measured results.

Table A1: Helmholtz coil calibration, representative data. Field read at the midpoint of the coil pair with the Hall magnetometer on the LOW FIELD range, zeroed at that point with the current off. The reading uncertainty is 0.010 mT on every point.

$I_{=}$ (A)	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.1	1.2
B_0 (mT)	0.42	0.82	1.24	1.66	2.09	2.51	2.92	3.34	3.79	4.21	4.60	5.03

Table A2: ESR resonance data for DPPH, representative data. Five repeat settings of the resonance current at each frequency, with the mean, the sample standard deviation, the standard error of the mean, and the field obtained from the calibration of Table 1.

Coil	ν (MHz)	I_1 (A)	I_2 (A)	I_3 (A)	I_4 (A)	I_5 (A)	$\bar{I}$ (A)	s_I (A)	$\sigma_{\bar{I}}$ (A)	B_0 (mT)	σ_{B_0} (mT)
large	15.0	0.119	0.114	0.113	0.116	0.123	0.1170	0.0041	0.0018	0.4767	0.0076
large	20.0	0.177	0.169	0.163	0.157	0.156	0.1644	0.0088	0.0039	0.6761	0.0165
large	25.0	0.211	0.211	0.214	0.203	0.203	0.2084	0.0051	0.0023	0.8612	0.0096
large	30.0	0.247	0.244	0.249	0.248	0.254	0.2484	0.0036	0.0016	1.0295	0.0069
medium	35.0	0.298	0.296	0.286	0.284	0.283	0.2894	0.0071	0.0032	1.2020	0.0133
medium	40.0	0.332	0.332	0.334	0.333	0.331	0.3324	0.0011	0.0005	1.3829	0.0021
medium	45.0	0.372	0.376	0.373	0.372	0.382	0.3750	0.0042	0.0019	1.5621	0.0080
medium	50.0	0.419	0.418	0.413	0.412	0.424	0.4172	0.0049	0.0022	1.7397	0.0092
medium	55.0	0.464	0.466	0.458	0.460	0.460	0.4616	0.0033	0.0015	1.9265	0.0062
medium	60.0	0.493	0.506	0.502	0.498	0.496	0.4990	0.0051	0.0023	2.0839	0.0096
medium	65.0	0.541	0.543	0.540	0.551	0.546	0.5442	0.0044	0.0020	2.2740	0.0084
medium	70.0	0.582	0.592	0.593	0.592	0.591	0.5900	0.0045	0.0020	2.4667	0.0085
medium	75.0	0.626	0.631	0.636	0.629	0.634	0.6312	0.0040	0.0018	2.6401	0.0075
small	80.0	0.668	0.671	0.666	0.669	0.672	0.6692	0.0024	0.0011	2.7999	0.0045
small	85.0	0.712	0.713	0.717	0.715	0.694	0.7102	0.0093	0.0041	2.9724	0.0174
small	90.0	0.770	0.759	0.754	0.761	0.771	0.7630	0.0073	0.0033	3.1946	0.0138
small	95.0	0.789	0.802	0.811	0.794	0.798	0.7988	0.0083	0.0037	3.3452	0.0157
small	100.0	0.843	0.836	0.839	0.848	0.842	0.8416	0.0045	0.0020	3.5253	0.0085
small	105.0	0.896	0.888	0.886	0.874	0.879	0.8846	0.0085	0.0038	3.7062	0.0159
small	110.0	0.931	0.936	0.932	0.924	0.929	0.9304	0.0044	0.0020	3.8989	0.0083
small	115.0	0.972	0.968	0.976	0.975	0.969	0.9720	0.0035	0.0016	4.0739	0.0067
small	120.0	1.016	1.015	1.009	1.002	1.020	1.0124	0.0070	0.0031	4.2439	0.0132
small	125.0	1.054	1.051	1.050	1.042	1.055	1.0504	0.0051	0.0023	4.4038	0.0096
small	130.0	1.099	1.108	1.087	1.098	1.101	1.0986	0.0076	0.0034	4.6066	0.0142

Table A3: Electromagnet calibration, representative data. Both branches of the minor loop between 3.00 A and 4.00 A. Field uncertainty 0.0005 T and current uncertainty 0.005 A on every point.

Rising sweep			Falling sweep		
I (A)	B_0 (T)	σ_B (T)	I (A)	B_0 (T)	σ_B (T)
3.00	0.3798	0.0005	4.00	0.4550	0.0005
3.10	0.3876	0.0005	3.90	0.4483	0.0005
3.20	0.3962	0.0005	3.80	0.4423	0.0005
3.30	0.4041	0.0005	3.70	0.4358	0.0005
3.40	0.4118	0.0005	3.60	0.4297	0.0005
3.50	0.4192	0.0005	3.50	0.4213	0.0005
3.60	0.4275	0.0005	3.40	0.4153	0.0005
3.70	0.4337	0.0005	3.30	0.4083	0.0005
3.80	0.4412	0.0005	3.20	0.4012	0.0005
3.90	0.4480	0.0005	3.10	0.3938	0.0005
4.00	0.4550	0.0005	3.00	0.3850	0.0005

Table A4: NMR resonance data, representative data. Rising current sweep only. Fields are computed from the quadratic calibration of Table 4 with the full coefficient covariance propagated.

	glycerine, ^{1}H			PTFE, ^{19}F		
I (A)	ν (MHz)	B_0 (T)	σ_B (T)	ν (MHz)	B_0 (T)	σ_B (T)
3.00	16.14	0.3796	0.0006	15.28	0.3796	0.0006
3.10	16.57	0.3879	0.0005	15.53	0.3879	0.0005
3.20	16.89	0.3961	0.0005	15.81	0.3961	0.0005
3.30	17.21	0.4040	0.0005	16.28	0.4040	0.0005
3.40	17.53	0.4118	0.0005	16.59	0.4118	0.0005
3.50	17.85	0.4194	0.0005	16.90	0.4194	0.0005
3.60	18.16	0.4269	0.0005	17.20	0.4269	0.0005
3.70	18.48	0.4341	0.0004	17.35	0.4341	0.0004
3.80	18.78	0.4412	0.0004	17.65	0.4412	0.0004
3.90	19.12	0.4481	0.0005	17.86	0.4481	0.0005
4.00	19.37	0.4549	0.0006	18.31	0.4549	0.0006
Frequency uncertainty, glycerine: 0.020 MHz				PTFE: 0.050 MHz		